

Amir Khajavirad

Edmonton, AB | Open to Relocation

+1 647 863 9030 | khajavir@ualberta.ca

linkedin.com/in/amirkhajavirad | github.com/Amirkhajavirad | amirkhajavirad.github.io

Summary

Biomedical Engineer focused on robotics integration, embedded systems, data acquisition, and experimental platform development. Experienced in building and debugging Raspberry Pi based robotic systems that combine sensors, motors, GUI controls, and real time data workflows. Strong background in system reliability, hardware software integration, experimental validation, and translating noisy real world system behavior into reliable and repeatable results.

Technical Skills

Programming and Data: Python, SQL, Bash, Linux

Embedded and Robotics Systems: Raspberry Pi, Embedded Systems, Sensor Integration, Motor Control, Serial Communication

Systems Integration: Hardware Software Integration, System Debugging, Calibration, Data Acquisition, Control Systems

Software and Interfaces: Object Oriented Programming, GUI Development, Touchscreen Interfaces, Git

Signal and Data Analysis: Filtering, Noise Reduction, Feature Extraction, Sensor Signal Interpretation, Experimental Design

Machine Learning: Scikit learn, XGBoost, SHAP, Feature Engineering, Model Optimization

Engineering and Design: SolidWorks, CAD Modeling, Mechanical Design, Prototyping

Mathematics: Linear Algebra, Probability, Statistics

Experience

Robotics Engineer Intern

Nov 2025 – Present

University of Alberta — Edmonton, AB

- Developed, tested, and improved a Raspberry Pi based robotic training system used for rodent rehabilitation experiments
- Integrated touchscreen GUI functions with robot control code to support robot ID selection, system monitoring, restriction settings, and operator commands
- Debugged hardware and software interactions across sensors, motors, RFID reader, load cell, pellet detection systems, and the main control program
- Troubleshoot real lab issues including calibration failures, PCB instability, wiring problems, USB communication reliability, serial port differences, and sensor alignment
- Built data acquisition workflows for multi modal sensor data including RFID, photo interrupt sensors, and load cell signals
- Designed and implemented classification workflows for event detection, outcome tracking, and system performance evaluation
- Improved system stability and reduced failure cases by 30 percent through debugging, calibration, and refinement of control logic
- Reduced system errors by 25 percent through hardware software testing, sensor alignment, and reliability focused troubleshooting
- Automated experimental workflows, reducing manual intervention by 60 to 70 percent and enabling repeatable data collection
- Developed TCP based Master Slave communication logic for synchronized dual robot operation and centralized session tracking
- Achieved real time system response with latency below 100 milliseconds for sensing and control tasks
- Built and validated electromechanical prototypes to support reliable experimental data acquisition

Projects

Standardized Biomechanical Testing Platform for Distal Biceps Fixation

- Designed and built a biomechanical testing platform for controlled mechanical data collection and performance analysis
- Developed 3D printed anatomical bone models and custom fixtures integrated with a BOSE ElectroForce mechanical testing system

- Implemented cyclic loading and pull to failure protocols based on biomechanical literature to evaluate fixation performance
- Engineered reproducible testing workflows to reduce experimental variability and improve consistency of results
- Enabled quantitative comparison of fixation methods through controlled data acquisition and mechanical performance analysis

PredictIQ Football Match Outcome Prediction (XGBoost)

github.com/shervinafzali/PredictIQ

- Developed an end to end machine learning pipeline including data extraction, feature engineering, model training, and evaluation
- Designed time aware validation strategies to prevent data leakage and simulate real world prediction conditions
- Optimized an XGBoost model achieving a Macro F1 score of 0.357 and outperforming baseline models
- Applied SHAP to interpret model behavior and communicate feature importance

Breast Cancer Subtype Classification

- Built classification models on high dimensional gene expression data with over 54000 features
- Applied dimensionality reduction techniques including PCA, Kernel PCA, and LDA to improve model robustness
- Evaluated multiple models including SVM, Random Forest, KNN, and Logistic Regression
- Performed outlier detection and preprocessing to improve data quality and reliability of analysis

Adaptive Circadian Lighting System for Neonatal Incubators

- Designed and implemented an embedded biomedical system for circadian aligned lighting control
- Developed microcontroller based control logic and integrated electronic components for dynamic light regulation
- Built and validated a complete hardware prototype through testing, debugging, and performance checks

Education

Master of Science, Biomedical Engineering

Jan 2024 – Dec 2025

University of Alberta — Edmonton, AB

- Developed a biomechanical testing platform integrating CAD design, 3D printed models, mechanical fixtures, and controlled testing workflows

Bachelor of Science, Biomedical Engineering

Sep 2017 – Jun 2022

Islamic Azad University — Tehran, Iran

- Designed and built an embedded biomedical device for neonatal circadian lighting control

Additional Experience

Recreation Facility Operator

Sep 2024 – Present

University of Alberta — Edmonton, AB

- Managed operations in a high traffic environment, ensuring safety, reliability, and real time issue resolution
- Maintained operational consistency under dynamic conditions while communicating clearly with staff and users

Leadership and Activities

- DRA Research Supervisor, University of Alberta Supervised undergraduate research activities, supported experimental setup development, and guided system level problem solving
- Event Coordination Volunteer, Engineering Graduate Research Symposium, University of Alberta
- Event Volunteer, BioAlberta Health and Life Sciences Showcase 2025